

RGI explained through local response

Shared batch difficulty dominates fluctuations; measured learning response changes with the optimizer.

$$L_t(B) = D(B) + A_t(B) + Q_{2,t}(B) + \varepsilon_t(B)$$

Same true prefixes and targets; frozen continuation start θ_0 ; $\Delta z = z_t - z_0$.

Full window: 512 updates. Ranges span three permutations of one pool, not confidence intervals. Unit: nats.

Term / statistic		How it is measured	SGD	Adam
<i>D</i>	Frozen difficulty	$\ell_{\theta_0}(B)$		
batch standard deviation		Frozen model cross-entropy on the same batch.	0.1843 – 0.1937	0.1843 – 0.1936
<i>A</i>	Linear work	$g_B(\theta_0)^\top (\theta_t - \theta_0)$		
mean over batches		Frozen batch gradient times actual parameter displacement.	−0.004011 – −0.003552	−0.003290 – −0.003272
<i>Q</i> ₂	Softmax curvature	$\frac{1}{2} \text{mean}_{i \in B} \text{Var}_{p_{0,i}}(\Delta z_i)$		
mean over batches		Full 50,304-token vocabulary; average across target positions.	+0.001744 – +0.001992	+0.000622 – +0.000634
ε	Measured remainder	$L - D - A - Q_2 = R + (Q - Q_2)$		
RMS over batches		Includes representation nonlinearity and curvature approximation.	0.0001909 – 0.0002863	0.0000754 – 0.0000838

Raw fluctuations	Common frozen checkpoint explains 96.66–97.58% of arriving-loss variance (near/far).
Local response	Negative <i>A</i> lowers loss; positive <i>Q</i> ₂ offsets part of the improvement. <i>A</i> + <i>Q</i> ₂ relative RMS error: SGD 4.97–9.73%; Adam 1.19–2.82% (near/far/full).
Validation boundary	Paired optimizer-gap error 10.34–16.68% exceeds the 10% gate. Universal strong RGI remains unproved.